

Original Article

Growth rates and microcystin production of *Microcystis* strains in response to allochthonous and autochthonous nutrient sources

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Abstract

This study investigated the growth dynamics and microcystin (MC) production of *Microcystis* in response to various autochthonous and allochthonous nutrient sources, specifically cyanobacterial scum, floating periphyton mats, sediment, and agricultural soil. Chemical analysis revealed that cyanobacterial scum is a major nitrogen reservoir (25.72 mg/g of dry weight(g dw)), while periphyton mats serve as significant phosphorus buffers through luxury uptake (1.92 mg/g dw). Laboratory cultures of *Microcystis* strains NIER10004 and CBE51 demonstrated that sediment extracts promoted the highest growth rates (0.0160 and 0.0166 OD₇₅₀/d, respectively) and the greatest biomass-to-total phosphorus ratios. Phosphorus uptake followed a biphasic pattern: a rapid depletion of orthophosphate in the first 10 days, followed by a transition to organic phosphorus utilization, likely facilitated by alkaline phosphatase activity. MC concentrations peaked at the onset of the stationary phase, though total MC content showed a low correlation with the specific nutrient source, suggesting that nutrient supply primarily drives biomass expansion rather than MC biosynthesis rates. Given the feasibility of mechanical removal, artificial harvesting of periphyton mats during spring—when increased microbial activity in sediments causes the mats to detach and float—is proposed as a more viable and cost-effective management strategy than expensive sediment dredging for mitigating harmful algal blooms in eutrophic reservoirs.

Keywords: Nutrient sources, Organic phosphorus, Harmful algal blooms, Periphyton mat, *Microcystis*, Microcystins

Introduction

Global temperature rises driven by climate change, coupled with increased organic matter loading from industrialization and urbanization, are accelerating the eutrophication of freshwater ecosystems worldwide [1]. Under these changing environmental conditions, harmful algal blooms (HABs) caused by cyanobacteria have emerged as a critical environmental issue, leading to aquatic oxygen depletion, loss of biodiversity, and disruption of ecosystem functions [2,3]. In particular, cyanobacterial genera such as *Microcystis*, *Anabaena*, and *Oscillatoria* pose a direct threat to the safety of drinking water sources by producing hepatotoxins like microcystins (MCs) and neurotoxins like anatoxins, resulting in exponentially increasing socio-economic costs [4,5]. In most reservoirs used as water sources in South Korea, HABs recur annually during the summer season [6].

The growth of cyanobacteria is influenced by a complex interplay of physical, chemical, and biological factors. Physically, as thermal stratification develops due to rising water temperatures, cyanobacteria utilize gas vacuoles to regulate buoyancy, allowing them to preemptively occupy optimal vertical positions for photosynthesis [7,8]. Biologically, they possess the ability to form colonies that resist predation by zooplankton [9,10]. However, the core mechanism determining the occurrence of HABs is the availability of chemical nutrients, particularly nitrogen (N) and phosphorus (P) [11,12].

The abnormal proliferation of cyanobacteria in freshwater ecosystems is closely linked to the mass balance of nutrients within the water body. Phosphorus, in particular, is the key limiting nutrient for freshwater cyanobacteria due to its essential role in cellular energy (ATP) and genetic material (DNA/RNA) synthesis [13]. When phosphorus concentrations exceed a critical threshold, cyanobacteria exhibit rapid metabolic activation, often leading to the dominance of specific species capable of nitrogen fixation [14,15]. Nitrogen is also a vital element for protein and nucleic acid synthesis; concentrations of ammonia nitrogen and nitrate nitrogen in the water act as key fuels that accelerate growth rates [16]. Furthermore, recent studies have demonstrated that not only the absolute concentrations of N and P but also their supply ratio (N:P ratio) significantly influence species composition and the intensity of toxin expression [17,18].

Cyanobacteria possess unique phosphorus utilization strategies. First, when the concentration of soluble reactive phosphorus (SRP) rises, cyanobacteria immediately absorb and store it as intracellular polyphosphate - a mechanism known as luxury consumption [19]. This internal storage provides the physiological resilience necessary to sustain growth even under P-limiting conditions. Second, phosphorus concentration is directly correlated with toxin gene expression. Molecular research suggests that when phosphorus levels exceed a specific threshold, the activity of *mcy* gene clusters increases, accelerating the production of MCs [20,21]. Third, phosphorus levels are amplified by internal loading, a dynamic interaction with sediments. The decomposition of dead organic matter during massive blooms depletes dissolved oxygen in the bottom layer, inducing a drop in redox potential and the release of phosphate bound to iron (Fe-P) back into the water column [22].

Nutrient loading into reservoirs is categorized into external and internal loads based on their origin. External loads, entering through point and non-point sources, are traditionally managed via the expansion of environmental infrastructure and the operation of reduction facilities [23,24]. Internal loads, originating from contaminated sediments, are typically controlled through dredging or the suppression of release via hypolimnetic aeration [25,26]. However, contemporary research suggests that these conventional approaches often fail to account for the complex biogeochemical cycling of phosphorus within the water column [27].

Crucially, dissolved inorganic phosphorus (DIP), which is directly available for biological uptake, often accounts for a mere fraction (approximately 5%) of the total phosphorus (TP) in many freshwater systems. The vast majority of phosphorus exists as organic phosphorus (OP), much of which is associated with biological debris and autochthonous organic matter [28,29]. Recent molecular evidence highlights that the bioavailability of this organic pool to cyanobacteria is highly variable and dependent on the specific chemical structure of the OP compounds [13]. Cyanobacteria have evolved sophisticated enzymatic mechanisms, such as the secretion of alkaline phosphatase (APase), allowing them to mineralize complex organic P forms into bioavailable inorganic P under P-limiting conditions [18,30].

Furthermore, existing control strategies that overlook nutrient cycling resulting from biological mortality - the "internal recycling" within the pelagic zone - have significant limitations in reducing the intensity and frequency of blooms [31,32]. Biological debris serves as a critical reservoir for not only phosphorus but also essential trace elements and organic ligands that can stimulate cyanobacterial growth even when external inputs are minimized [31,33]. Therefore, understanding the legacy of nutrients derived from biological decay is essential for developing next-generation restoration strategies that can effectively mitigate persistent HABs in a warming climate.

Consequently, there is a need for precise analysis regarding the impact of biologically derived nutrients on cyanobacterial growth. Therefore, this study aims to analyze the growth rates and biomass changes of cyanobacteria in response to substances introduced from various nutrient sources both within and outside reservoirs, including agricultural soil, sediment, cyanobacterial scum, and floating periphyton mats.

Materials and Methods

Cultivation of cyanobacteria

To investigate the growth dynamics of cyanobacteria in response to different nutrient sources, two pure-isolated strains of the genus *Microcystis* were utilized. *M. aeruginosa* NIER10004, isolated from Soyang Reservoir, was provided by the National Institute of Environmental Research (NIER), and *Microcystis* sp. CBE51 was isolated from Chungju Reservoir. The strains were cultured in BG-11 medium [34]. Cultivation was performed in an incubator at 25 °C with an irradiance of 40 $\mu\text{molE}/\text{m}^2/\text{s}$ and a stirring speed of 50 rpm, under a 14:10 hr light/dark cycle.

Sampling and pretreatment of nutrient sources

Four types of nutrient sources were used in this study: three autochthonous sources (floating periphyton mats, cyanobacterial scum, and sediment) and one allochthonous source (agricultural soil). Floating periphyton mats were collected from Geori, Daechung Reservoir (36°20'18" N, 127°33'54" E), and sediments were sampled from the upper reaches of Seohwa Stream, Daechung Reservoir (36°21'33" N, 127°34'03" E). Cyanobacterial scum was collected from Dodongnaru on the Nakdong River (35°41'10" N, 128°22'19" E), and agricultural soil was sampled from Chusori, Daechung Reservoir (36°21'16" N, 127°33'33" E).

The floating periphyton mats were collected using plastic containers and separated from the reservoir water. They were dried at 50°C for three days, pulverized, and stored at -70°C. Sediments were collected from the upper layer using a grab sampler, mixed with a glass rod, dried at 50°C, pulverized, and stored at -70°C. For the cyanobacterial scum, the samples were kept in a dark room at 4°C for one week to allow the scum to float to the surface. The upper layer was then isolated, dried at 50°C, and stored at -70°C. Agricultural soil was prepared by removing roots and debris, followed by drying at 50°C and sieving through a 2 mm mesh to remove large particles and gravel before storage.

Analysis of nutrient release from sources

To analyze the concentration of releasable nutrients, each sample was added to a P-free BG-11 medium (BG-11 medium deficient in potassium phosphate dibasic, K_2HPO_4) and subjected to pretreatment. For sediments, nutrients were extracted using an ultrasonic processor (KFS-150N, Hankukoprotech, Korea) at 80W for 30 sec (repeated 5 times), followed by stirring at 150 rpm for 1 hr at room temperature. The pretreated sediments were autoclaved at 121°C for 30 min, left to settle for 24 hr, and the supernatant was collected for analysis. To simulate the natural mortality and decomposition of cyanobacterial scum and periphyton mats, cells were lysed via ultrasonication (80 W, 30 sec $\times$ 5 times), stirred for 1 hr at room temperature, and the supernatant was collected after 24 hr. For agricultural soil, which typically enters reservoirs via rainfall runoff, the samples were stirred for 1 hr and left for 24 hr at room temperature without additional lysis before the supernatant was collected.

Concentrations of total nitrogen (TN), dissolved TN (DTN), TP, dissolved TP (DTP), and orthophosphate (PO_4^{3-} -P) were analyzed according to the Standard Methods for the Examination of Water Pollution [35].

Cyanobacterial growth by nutrient sources

To compare growth rates, pretreated nutrient extracts were added to P-free BG-11 medium to achieve a final TP concentration of 0.5 mg/L. K_2HPO_4 was used for the control group. The pH of the culture solutions was adjusted to 7.10 (± 0.05) using HCl or NaOH. Prior to inoculation, the two *Microcystis* strains were centrifuged and washed to prevent shifts in TP concentration.

Cultivation conditions were maintained at 25°C 40 μ molE/m²/s, 50 rpm, and a 14:10 hr light/dark cycle. Samples were collected every 3 to 5 days for physicochemical, biomass, and MCs analysis. Biomass was measured via absorbance at 750 nm using a spectrophotometer (Optizen 2120UV; Mecasys, Korea), a standard proxy for algal biomass in laboratory cultures [36]. Nutrient and chlorophyll-a (Chl-a) concentrations were analyzed following standard protocols [35].

Concentrations of microcystins

For MC analysis, samples were filtered through GF/C filters (Whatman, UK) and dried at 50°C for 1 hr. MCs were extracted by adding 2 mL of methanol and stirring for 1 hr at room temperature. The extract was filtered using a 0.2 μ m PTFE hydrophilic syringe filter (Advantec, Japan) and analyzed for MC-LR, -RR, and -YR concentrations using HPLC [37]. The HPLC system (Waters 1525) was equipped with a photodiode array detector (Waters 2998) and an autosampler (Waters 2707), with data processed via Empower Pro software. An Xterra RP18 column (5 μ m particle size, 15 cm $\times$ 3.9 mm I.D.; Waters) was used. The mobile phase consisted of acetonitrile and water (68:32) containing 0.05% (v/v) trifluoroacetic acid at a flow rate of 0.8 mL/min. Standard MC-LR, -YR, and -RR (Sigma-Aldrich Co., USA) were used for quantification.

Statistical analysis

All experimental data were expressed as the mean $\pm$ standard deviation (SD) of triplicate measurements. To evaluate the statistical significance of differences in growth rates and biomass-to-TP ratios (OD_{750}/TP) among various nutrient sources, a one-way analysis of variance (ANOVA) was performed. Prior to the ANOVA, the normality of the data and homogeneity of variance were verified using Shapiro-Wilk and Levene's tests, respectively [38]. When significant differences were detected ($P < 0.05$), Tukey's Honestly Significant Difference (HSD) post-hoc test was applied for multiple comparisons to identify specific differences between treatment groups. The stability of nutrient concentrations over the incubation period was assessed using linear regression analysis, where slopes and coefficients of variation (CV) were calculated to determine the rate of change [39]. Additionally, Pearson correlation analysis was conducted to evaluate the relationship between nutrient source types and MC production, including both total MCs and individual variants [40]. All statistical analyses were executed using R software (version 4.2.0).

Results

Nutrient content in nutrient sources

The nitrogen and phosphorus concentrations of the dried nutrient source samples were analyzed. Cyanobacterial scum exhibited the highest TN concentration at 25.72 mg/g of dry weight(gdw), followed by floating periphyton mats at 8.69 mg/gdw (Table 1). Regarding TP, floating periphyton mats showed the highest concentration at 1.92 mg/gdw, while cyanobacterial scum recorded 1.66 mg/gdw. The TN/TP ratios were 15.47 for cyanobacterial scum and 4.53 and 6.89 for floating periphyton mats and sediments, respectively. The intracellular N:P ratio of 16:1, as proposed by Redfield, serves as an indicator of whether nitrogen or phosphorus acts as the limiting factor [41]. It has also been noted that intracellular N:P ratios can vary by more than twofold depending on the ambient nutrient concentrations in the water body [42]. While the TN/TP ratio of cyanobacterial scum was similar to the Redfield ratio, the periphyton mats showed a significantly lower value. This low ratio in periphyton mats likely results from luxury uptake of phosphorus during P-cycling between the

sediment and the water column. Previous research by Oh & Cho reported that organic matter derived from cyanobacterial blooms is distributed in the sediments of the Chuso-ri area in Daechung Reservoir, with sediment TN/TP ratios ranging from 7.99 to 8.77 [43].

Table 1. Nitrogen and phosphorus concentrations of the nutrient sources.

Nutrient sources	Concentration (mg/g of dry weight)				
	Total nitrogen	Dissolved total nitrogen	Total phosphorus	Dissolved total phosphorus	Orthophosphate
Periphyton mat	8.69	1.29	1.92	0.50	0.44
Cyanobacterial scum	25.72	20.79	1.66	0.61	0.53
Sediment	0.52	0.40	0.076	0.038	0.030
Agricultural soil	N.A. ¹	N.A.	0.12	N.A.	N.A.

¹ N.A.: Not Analyzed

Cyanobacterial growth dynamics by nutrient source

Biomass changes in response to different nutrient sources were monitored in laboratory-scale cultures of two *Microcystis* strains (Fig. 1). For the NIER10004 strain, the highest OD₇₅₀ values were observed on Day 26 in cultures supplemented with lysate from cyanobacterial scum and sediment extract, while the periphyton mat extract reached its peak on Day 17. For the CBE51 strain, OD₇₅₀ peaked on Day 44, with the sediment extract yielding the highest value (0.623), followed by cyanobacterial lysate (0.608) and periphyton mats (0.516).

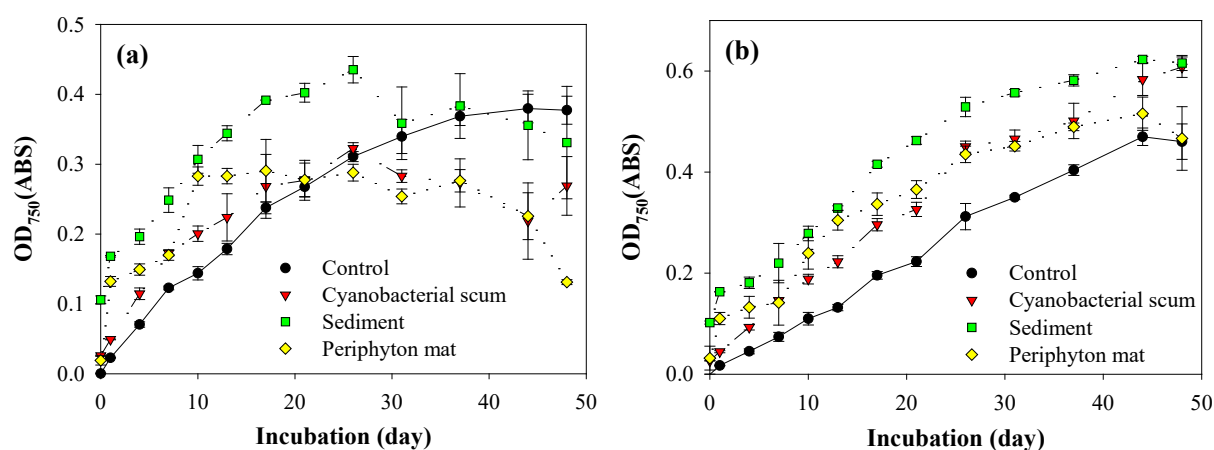


Figure 1. Comparison of *Microcystis* biomass across different nutrient sources in laboratory-scale experiments: (a) NIER10004; (b) CBE51. Error bars indicate the standard deviation of triplicate samples.

The NIER10004 strain transitioned to the stationary phase between Days 17 and 21 in the cyanobacterial lysate and sediment treatments, while the periphyton mat treatment plateaued after an initial 10-day growth period. Biomass began to decline across all nutrient sources after Day 31. The CBE51 strain reached its maximum growth rate on Day 26. Analysis of the biomass change between Days 26 and 32 showed reductions of 76.0% in the cyanobacterial lysate treatment, 68.9% in the sediment extract, and 54.0% in the periphyton mat treatment.

Growth rate analysis revealed that both strains achieved their highest growth rates in the sediment extract: 0.0160 OD₇₅₀/d for NIER10004 (n=18) and 0.0166 OD₇₅₀/d for CBE51 (n=24). These were followed by periphyton mat extract (0.0148 and 0.0156, respectively) and cyanobacterial lysate (0.0141 and 0.0146, respectively). Univariate analysis yielded F-values of 69.02 for NIER10004 and 119.58 for CBE51, indicating significant differences in growth rates depending on the nutrient source (P<0.01). Tukey's HSD post-hoc test confirmed significant differences across all sources (P<0.05), with the highest growth rates in sediment extract, followed by periphyton mat extract and cyanobacterial lysate.

To determine whether the type of nutrient source significantly influenced cyanobacterial biomass dynamics, the ratio of OD₇₅₀ to TP concentration in each culture vessel was calculated and subjected to ANOVA (Table 2). For the NIER10004 strain, the ANOVA yielded an F-value of 13.627. Subsequent Tukey's HSD post-hoc test categorized the nutrient sources into two distinct groups: the first comprising cyanobacterial lysate and floating periphyton mat extract, and the second consisting of sediment extract. Notably, the biomass increase per unit TP was significantly higher in the sediment

extract treatment. For the CBE51 strain, the treatments were differentiated into three separate groups. The highest biomass increases were observed in the cultures supplemented with cyanobacterial lysate and floating periphyton mat extract. Synthesizing the results for growth rates and biomass yield per unit TP, both *Microcystis* species exhibited the most robust growth and the highest biomass-to-TP increase ratio in the sediment extract treatment. In the CBE51 cultures, the growth rate and biomass yield per unit TP showed identical trends across the different nutrient sources. In contrast, while the NIER10004 strain showed significant variations in growth rates depending on the nutrient source, the biomass-to-TP ratio remained statistically consistent across all sources except for the sediment extract.

Table 2. One-way ANOVA of the biomass-to-total phosphorus ratio (OD_{750}/TP) across nutrient sources. (a) control; (b) cyanobacterial scum; (c) sediment; (d) periphyton mat.

Strain	n	F value	P	Tukey post hoc test ($P < 0.05$)
<i>Microcystis aeruginosa</i> NIER10004	36	13.627	<0.01	abd < c
<i>Microcystis</i> sp. CBE51	36	7.620	<0.01	ad < bd < bc

¹ (a) control; (b) cyanobacterial scum; (c) sediment; (d) periphyton mat

Nitrogen and phosphorus concentrations in incubation

Fig. 2 illustrates the changes in TN concentrations. For NIER10004, the average TN concentrations were 70.9 (± 2.1) mg/L (cyanobacterial lysate), 68.3 (± 2.8) mg/L (sediment), and 68.7 (± 2.6) mg/L (periphyton mats). For CBE51, concentrations were 70.9 (± 2.2) mg/L (cyanobacterial lysate), 67.2 (± 1.9) mg/L (sediment), and 67.9 (± 2.3) mg/L (periphyton mats). Analysis of TN concentrations over time showed CV ranging from 2.83% to 3.67% for the NIER10004 strain and 2.67% to 3.31% for the CBE51 strain, indicating that TN concentrations remained statistically stable throughout the laboratory-scale incubation period.

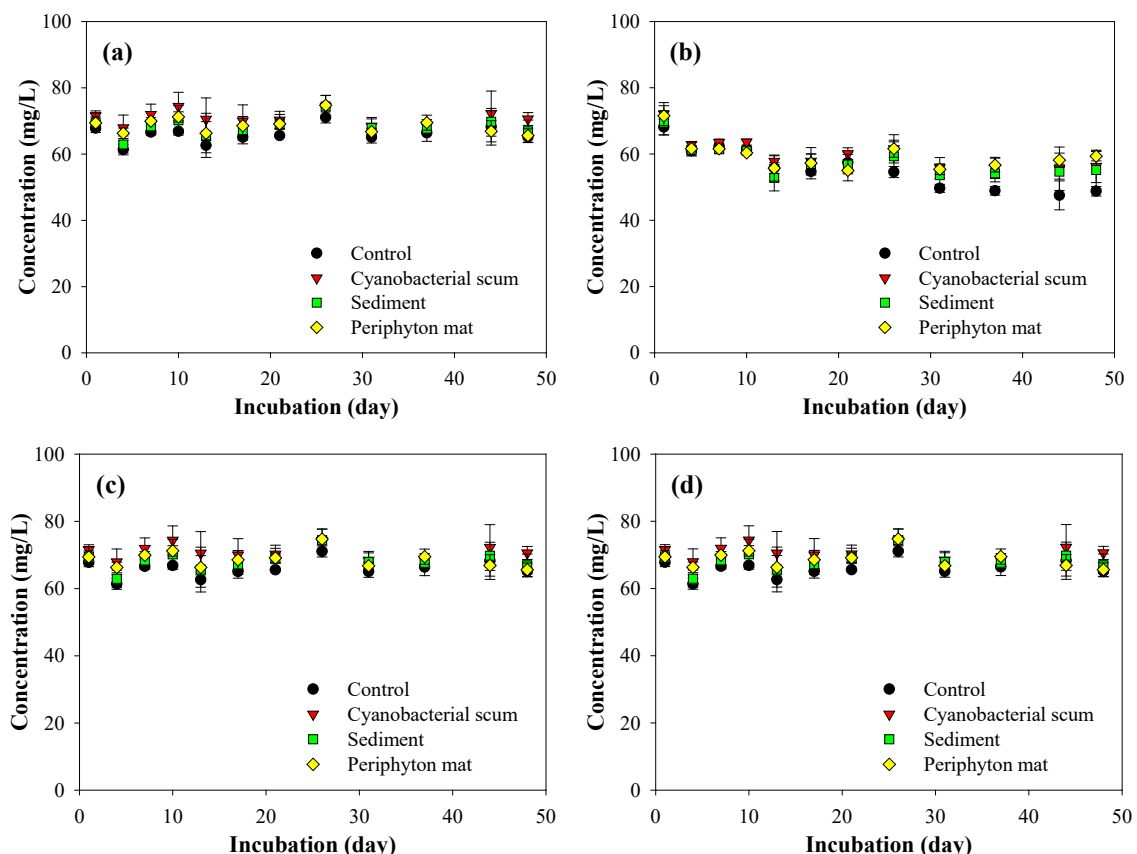


Figure 2. Variations in nitrogen concentrations during laboratory-scale experiments: (a) total nitrogen - NIER10004; (b) dissolved total nitrogen - NIER10004; (c) total nitrogen - CBE51; (d) dissolved total nitrogen - CBE51. Error bars indicate standard deviations of triplicate samples.

In contrast, DTN concentrations consistently decreased from an initial 70.2 (± 1.5) mg/L to a final 54.9 (± 4.5) mg/L.

In the NIER10004 cultures supplemented with periphyton mat extract, the DTN concentration decreased by 23.1% (from 71.6 mg/L to 55.1 mg/L), with similar reductions observed in the cyanobacterial lysate (22.9%) and sediment extract (24.0%) treatments, indicating a consistent decline across all nutrient sources. In the CBE51 cultures, the DTN concentration in the periphyton mat treatment decreased by 23.7% (from 69.8 mg/L to 53.2 mg/L), while the cyanobacterial lysate and sediment extract treatments showed greater reductions of 34.9% and 30.4%, respectively. These trends suggest that dissolved nitrogen was assimilated for cellular synthesis during *Microcystis* growth and subsequently converted into particulate nitrogen.

The TP concentrations for CBE51 cultures remained stable, whereas a decrease was observed in NIER10004 cultures over the incubation period (Fig. 3). The average TP concentrations in the NIER10004 cultures were 0.58 mg/L for cyanobacterial lysate, 0.63 mg/L for sediment extract, and 0.71 mg/L for the periphyton mat treatment. In contrast, the CBE51 cultures showed average TP concentrations of 0.72 mg/L, 0.72 mg/L, and 0.81 mg/L for the cyanobacterial lysate, sediment extract, and periphyton mat treatments, respectively. The calculated rate of change in TP concentration over time ranged from -0.0021 to -0.0081 mg/L/d for NIER10004, while for CBE51, it remained between 0.00016 and 0.00092 mg/L/d. Unlike the CBE51 strain, which grew by forming surface scums, the NIER10004 strain did not form scums and was observed to grow while attached to the vessel walls. Additionally, after 30 days of incubation, both dead and viable cells were found adhered to the bottom of the culture vessels. Therefore, the observed decrease in TP concentration in the NIER10004 cultures is attributed to the exclusion of these wall-attached and settled *Microcystis* cells during the sampling process.

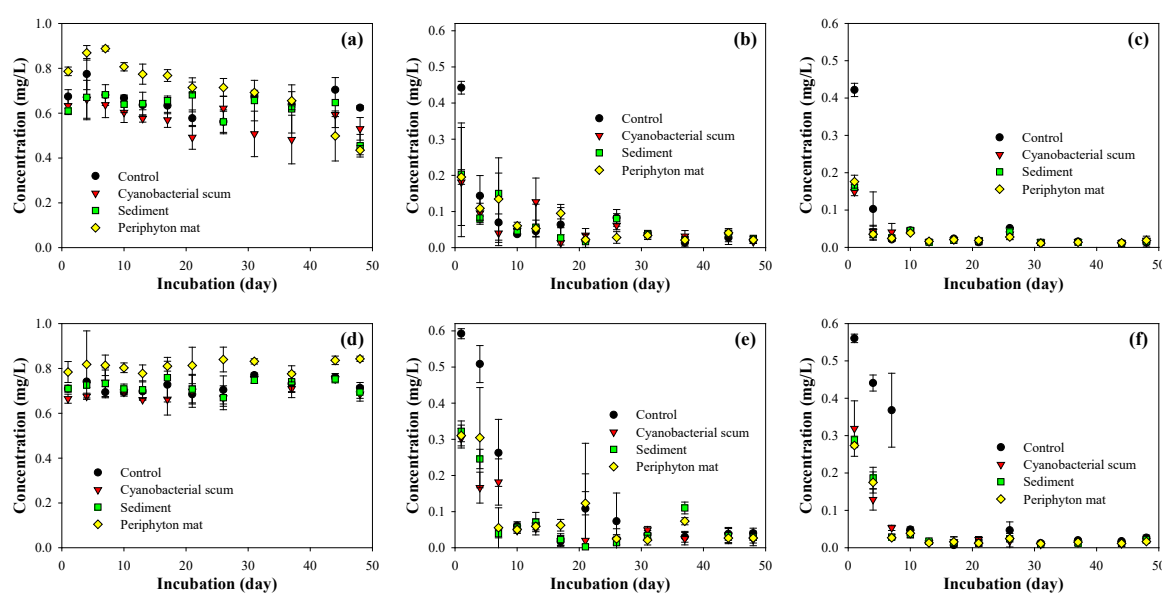


Figure 3. Variations in phosphorus concentrations during laboratory-scale experiments. (a) total phosphorus - NIER10004; (b) dissolved total phosphorus - NIER10004; (c) orthophosphate - NIER10004; (d) total phosphorus - CBE51; (e) dissolved total phosphorus - CBE51; (f) orthophosphate - CBE51. Error bars indicate standard deviations of triplicate samples.

Analysis of DTP and orthophosphate concentrations revealed a sharp decline in concentrations across all culture vessels during the first 10 days of incubation. Following this initial period, DTP concentrations were maintained between 0.027 and 0.040 mg/L, while orthophosphate levels remained within the range of 0.012 to 0.018 mg/L, regardless of the inoculated strain or nutrient source. Cyanobacteria of the genus *Microcystis* are known to preferentially utilize orthophosphate from the water column before consuming organic phosphorus via APase activity [44]. Given the rapid decrease in dissolved phosphorus observed in Fig. 3, it is inferred that the strains initially assimilated orthophosphate and subsequently transitioned to the uptake of OP components as the inorganic pool became depleted.

Microcystins analysis

The NIER10004 strain produced MC-LR and -RR, while CBE51 produced MC-LR, -RR, and -YR. In both strains, MC concentrations increased initially before declining (Fig. 4). For NIER10004, peak concentrations occurred on Day 17 in the periphyton mat (1.89 µg/L) and cyanobacterial lysate (1.54 µg/L) treatments. The CBE51 strain reached its total MC peak on Day 31 across all sources, with the highest concentration in the cyanobacterial lysate (1.18 µg/L), followed by sediment (0.80 µg/L) and periphyton mats (0.61 µg/L). MC-YR became undetectable in CBE51 cultures after Day 31 or 37, depending on the source. The ratio of MCs to biomass (MCs/OD₇₅₀) peaked concurrently with the maximum MC concentrations. In all treatments, the highest MC content per unit of biomass was observed at the onset of the stationary phase. Correlation

analysis showed significant differences in MC production depending on the nutrient source ($P < 0.01$; Table 3). However, Tukey's post-hoc test for NIER10004 indicated that MC concentrations were actually higher in the control group than in the nutrient source treatments. For CBE51, the cyanobacterial lysate treatment yielded the highest MC levels. While some studies suggest that toxic *Microcystis* outcompetes non-toxic strains under high inorganic nitrogen conditions [45], the results of this study suggest that the specific type of nutrient source has a relatively low correlation with total MC content.

Discussion

Cyanobacteria exhibit a distinct ecological advantage by maintaining high APase activity compared to other algae or bacteria, allowing them to efficiently utilize OP even when DIP is limiting [13,46]. Previous studies have indicated that APase activity significantly increases when DIP concentrations fall below 1 mg $\text{PO}_4^{3-}\text{-P/L}$, whereas it remains suppressed above this threshold. Furthermore, under low phosphorus conditions (< 0.1 mg $\text{PO}_4^{3-}\text{-P/L}$), a positive correlation exists between dissolved OP concentrations and APase activity [47]. When inorganic phosphorus (IP) is scarce, *Microcystis* significantly upregulates genes such as *pstS* and *sphX* (by 50-400 times), which are involved in phosphorus uptake from phosphoproteins, alongside the APase-related *phoX* gene. This genetic plasticity provides *Microcystis* with a competitive edge for dominance in P-deficient environments [44].

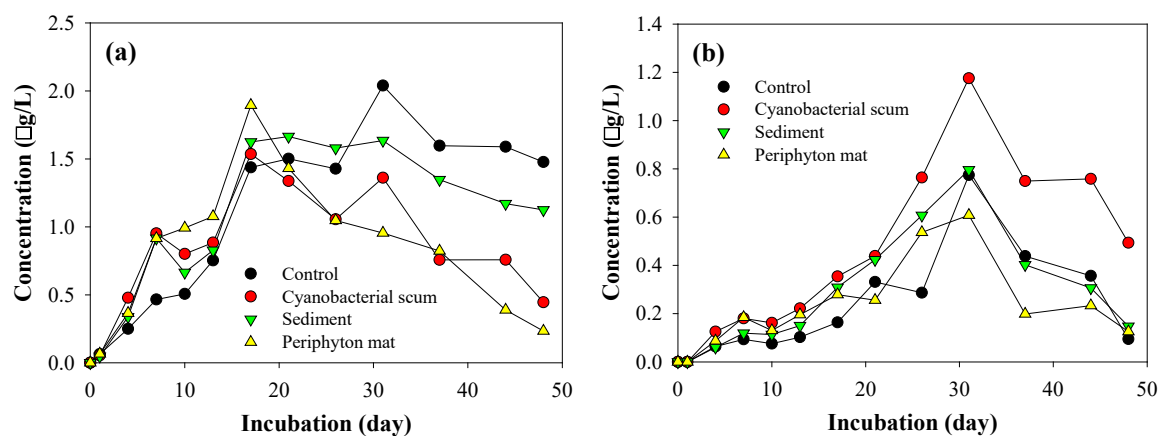


Figure 4. Time course of total intracellular microcystins according to nutrient sources. (a) NIER10004; (b) CBE51.

Table 3. One-way ANOVA of microcystin content per unit biomass (MCs/OD₇₅₀) according to nutrient source.

Strain	Microcystins	n	F value	P	Tukey post hoc test ($P < 0.05$)
NIER10004 ¹	MC-LR	36	6.863	< 0.01	bcd < a ²
	MC-RR	36	4.013	0.009	bcd < a
	Total MCs	36	9.817	0.01	bcd < a
CBE51	MC-LR	36	6.352	< 0.01	acd < ab
	MC-RR	36	6.828	< 0.01	cd < abd
	MC-YR	36	10.930	< 0.01	acd < b
	Total MCs	36	5.661	< 0.01	cd < ad < ab

¹ MC-YR is not detected in *Microcystis aeruginosa* NIER10004

² (a) control; (b) cyanobacterial scum; (c) sediment; (d) periphyton mat

Biological debris serves as a primary source of both OP and essential trace elements [48]. Consequently, current cyanobacterial control strategies that overlook nutrient cycling driven by biological mortality face inherent limitations in reducing bloom intensity and frequency. This necessitates an in-depth analysis of biogenic nutrients and their specific impacts on *Microcystis* growth. Experimental evidence shows that OP additions yield more pronounced growth-promoting effects than IP; for instance, final Chl-a concentrations in OP cultures (391.1 mg/m³) were substantially higher than those in IP cultures (250.3 mg/m³) [49]. Phycocyanin levels, a key cyanobacterial indicator, also reached 10,915 RFU in OP treatments, surpassing IP treatments (8,492 RFU). Notably, after day 30, phycocyanin levels plateaued or declined while Chl-a continued to rise, suggesting a taxonomic shift where green algae or diatoms became dominant by utilizing nutrients released after cyanobacterial senescence.

The chemical composition of nutrient sources is a critical determinant of *Microcystis* bloom dynamics. In this study,

cyanobacterial scum exhibited a TN concentration of 25.72 mg/gdw, approximately three times higher than that of floating periphyton mats, functioning as a concentrated nitrogen reservoir for subsequent blooms via mineralization. Conversely, periphyton mats recorded the highest TP concentration (1.92 mg/gdw) and a TN/TP ratio (4.53) far below the Redfield ratio (16:1). This suggests that periphyton communities at the sediment-water interface act as nutrient buffers through luxury phosphorus uptake [19,50]. Regarding toxicity, MC concentrations peaked at the onset of the stationary phase before declining. The relatively low correlation between the type of nutrient source and total MC content suggests that nutrient sources primarily influence biomass expansion rather than the specific rate of toxin biosynthesis. Thus, while nutrient supply determines the total toxin load in the water column, individual cellular toxin synthesis is governed by complex internal metabolic states and environmental stressors [51].

The DIP-to-DOP ratios of the nutrient sources analyzed in this study were 8:1 for periphyton, 7:1 for scum, and 4:1 for sediment. Given that the optimal inorganic-to-organic phosphorus ratio for *Microcystis* growth is reported as 6:1 [52], these sources significantly enhanced growth compared to IP alone. Since the half-saturation constant (K_s) of *M. aeruginosa* for phosphorus is as low as 0.024 mg/L [53], reducing reservoir phosphorus below this biological threshold is practically unfeasible. Therefore, preventing OP influx by removing nutrient-rich biogenic sources is a more effective strategy for mitigating blooms. While sediment removal is theoretically the primary countermeasure due to its high growth contribution, it is hindered by high costs and ecological disruption [54]. Periphyton mats attached to the surface layer of sediments detach in the form of scum and enter the water column due to gas bubbles generated by increased microbial activity in the sediment as water temperature rises, particularly in spring [55]. The floated periphyton mats are large and structurally robust, enabling easy removal by mechanical collectors while minimizing the use of chemical coagulants.

Consequently, the artificial removal of periphyton mats during the spring season emerges as the most viable and effective strategy for managing the frequency and intensity of blooms in eutrophic reservoirs.

Conclusions

In conclusion, this study demonstrates that the origin and chemical composition of nutrient sources are critical determinants of *Microcystis* bloom dynamics. The experimental results confirm that autochthonous organic matter, particularly sediment and cyanobacterial scum, functions as a significant internal reservoir of bioavailable nitrogen and phosphorus. Sediment extracts were identified as the most potent nutrient source, promoting the highest growth rates and biomass-to-TP ratios for both *Microcystis* strains.

The findings highlight a sophisticated resource acquisition strategy in which *Microcystis* preferentially utilizes IP before transitioning to OP pools via APase activity as inorganic levels become depleted. Furthermore, while nutrient availability dictates the overall biomass expansion and the resulting total MC load in the water column, the specific rate of MC biosynthesis appears to be governed by complex internal metabolic states rather than the nutrient source type alone.

From a management perspective, the study suggests that relying solely on external load reduction may be insufficient to control persistent harmful algal blooms. Given that the half-saturation constant for phosphorus in *M. aeruginosa* is exceptionally low, management efforts should prioritize the removal of internal nutrient-rich biogenic sources. Specifically, artificial harvesting of periphyton mats during the spring season—when increased microbial activity in sediments causes periphyton mats to detach and float—is proposed as the most viable and effective strategy for managing the frequency and intensity of blooms in eutrophic reservoirs, offering a practical alternative to the high costs and ecological disturbances associated with sediment dredging.

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Conflict of interest

The author declare there is no conflict of interest in this study. The author certifies that they have no known competing financial interests or personal ties that could have seemed to affect the work disclosed in this paper.

CRedit author statement

YJK: Conceptualization, Methodology, Analysis, Original Draft Preparation; KHO: Data Curation, Writing-Review & Editing; YCC: Supervision, Writing- Reviewing and Editing.

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